Enhancing Mental Health Prediction Using Advanced Large Language Models: A Multi-Platform, Systematic Evaluation Approach

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***Abstract*—One trillion people across the globe experience mental disorders as these conditions create more than fourteen percent of total disability which affects approximately one billion people globally (World Health Organization, 2019) ACE has created "Mental-LLM" based on Alpaca linked with FLAN-T5 to investigate text-based detection for early mental health assessment but this method faces limitations due to constrained architecture design along with one-site training and limited fairness tests (Xu et al., 2023), (Yang et al., 2023a). Our research combines eight publicly accessible corpora from Twitter alongside Facebook comments and empathetic dialogues and specialized forums when implementing a thorough twelve-step cleaning operation for HTML removal as well as slang expansion and emoji deletion and lemmatization.**

Meta-Llama 3.1-8B-Instruct-Instruct receives a 4-bit NF4 QLoRA model fine-tuning for a four-way balanced classification (Normal, Depression, Anxiety, Bipolar) via the execution of 4-bit NF4 QLoRA on a single platform. Our model with fine-tuning generates 78.4% macro-average accuracy which surpasses both the original base model by 9.1 points and Mental-LLM by 15.2 points while ensuring a two-second maximum latency for each sample analysis on 16 GB GPU infrastructure. The combination of state-top-tier LLMs with diverse data through parameter-efficient fine-tuning produces both excellent performance and low resource utilization therefore enabling practical mental-health screening systems for resource-limited areas.

***Index Terms*—QLoRA, NF4, MentaLLaMA, IMHI**

I. Introduction and Statement of the Problem

Mental health disorders stand as one of the biggest obstacles which affect global human welfare and medical wellness. Mental disorders caused 14.6 percent of disability in people worldwide in 2019 which translated into 125 million years lived with disability (YLDs) and resulted in about 16 percent of disability-adjusted life years (DALYs) thus demonstrating their extensive negative influence on the productivity of individuals and communities (Salomon et al., 2022), (Vos et al., 2021). Research indicates that depression generated more than one-third of YLD burden from mental disorders and both anxiety disorders and bipolar affective disorders became major disability causes worldwide. These health conditions cause extensive financial losses for society due to workplace absence combined with employees working through illness as well as illness-related wage loss from inadequate mental health treatment which costs the global economy more than US $1 trillion annually.

Proof-based treatment options including pharmacological therapy and psychotherapy along with community support programs are available yet fewer than 50 percent of mental disorder patients receive minimally appropriate care within high-income countries and this percentage reduces even lower in low- and middle-income nations to under 20 percent (National Alliance on Mental Illness, 2021). The average delay between symptom development and start of treatment lasts ten years or longer which causes deterioration in clinical results and boosts the chance of lasting psychiatric conditions. Healthcare facilities with personnel shortages experience bottlenecks because they must use professional in-person interviews to perform screening and diagnoses based on traditional pathways.

User-generated text data has become the subject of research concerning automated screening techniques that use large language models (LLMs). Through the "Mental-LLM" framework researchers showed that Alpaca and FLAN-T5 architectures trained in a fine-tuned manner can identify symptoms of mental disorders that manifest in online content. The first-generation systems operate with fundamental weaknesses that significantly impair their operational capabilities. These systems learn from exclusive platforms (for instance, Reddit alone) which results in limited abilities to identify diverse linguistic patterns that appear on Twitter and Facebook as well as specialized support forums. The models' basis stems from earlier technology than modern generative advancements while their post-training assessment for fairness remains incomplete so demographic preferences or systematic misclassification could occur (Xu et al., 2023) (Yang et al., 2023a).

Our research examines whether a 4-bit NF4 QLoRA fine-tune of Meta-Llama 3.1-8B-Instruct-Instruct with its 8-billion parameters can reach higher accuracy and lowered bias through its deployment on common GPUs which possess 16 GB of VRAM when processing diverse multilevel textual data (Dettmers et al., 2023). Our research collects eight diverse public datasets through a 12-step text-cleaning pipeline to

provide an extensive evaluation against base-LLM freezing and published Mental-LLM results for showing a deployable method to achieve accurate mental-health screening with reduced biases.

II. Review of Literature

The automated identification of mental-health indicators present in social media has made substantial progress throughout the last ten years. Research by De Choudhury et al. conducted depression classification using support vector machines (SVM) on lexicon and behavioral features from Twitter timelines which generated findings in the low eighties for accuracy rates (De Choudhury et al., 2013). BERT emerged as the standard text classification model due to its deep contextual embedding capabilities thus allowing researchers to differentiate #MentalHealth and #Depression tweets with greater than 80% accuracy when reviewing COVID-19 pandemic social media content (Panchal et al., 2021). The latest prompt-based zero- and few-shot systems that use GPT-3 and ChatGPT demonstrate potential by generating understandable explanations next to predictions but they presently achieve subpar classification results compared to domain-specific models (Martin, 2024).

Through the IMHI dataset MentaLLaMA created a 105 k-sample instruction set from various sources to optimize LLaMA2 performance and develop detailed mental-health evaluations which represent the initial release of LLMs for social media interpretation (Yang et al., 2023b). The efficiency of QLoRA enables full optimization of multi-billion-parameter LLMs by utilizing both 4-bit NF4 quantized weights and low-rank adapters which deliver identical downstream performance on a consumer-grade GPU (Greenslade, 2023). No prior research exists which benchmarks the Meta-Llama family against multi-platform mental-health classification responsibilities even though these instruction-tuned 8 B-parameter models represent some of the market's most advanced systems.

III. Objectives of the Study

- The first objective involves merging eight publicly accessible labeled datasets that include Twitter data along with Facebook comments and empathetic dialogues and specialized forums.
- Develop a unified, instruction-style prompt for four-way classification of Normal, Depression, Anxiety, and Bipolar.
- The research will conduct a 4-bit NF4 QLoRA retraining of Meta-Llama 3.1-8B-Instruct-Instruct followed by a performance benchmark against frozen Meta-Llama along with MentaLLaMA benchmarking results.
- The evaluation for fairness entails reporting both confusion-matrix shifts alongside disparity metrics while conducting model fine-tuning within 16 GB GPU memory constraints.

IV. Data Collection

We gathered eight separate public data collections for tracking various mental health expressions in written text. More than 500 000 Twitter mental-health tweets exist in the Mental-Disorder Tweets & Music dataset which has multiple disorder labels (Enamdari, 2024). The Twitter #MHCampaign Sentiment collection contains 21 000 tweets which have been tagged for sentiment analysis (Garg, 2024) but the Mental-Health Corpus presents 30 000 posts along with four main disorder categories (Facebook, 2023). The CAMS repository from GitHub includes 11 000 forum entries that are suicide risk annotated while the Depression-Anxiety Facebook Comments dataset contains 135 000 user comments (Jamal, 2024) (University of North Texas, 2025). Researchers utilize Empathetic Dialogues which contains 25 000 examples together with ESConv with its 9 000 examples to provide detailed emotion and strategic information. The multi-platform corpus contains 4 000 000 Twitter posts found within a private repository (Bal, 2020a). The project unified all CSV/JSON raw files through validation steps followed by concatenation to create the cleaned master CSV file.

V. Exploratory Data Analysis (EDA) and Hypotheses for the Study

Our analysis of the merged data collection focused on both quality assessment and pattern recognition that would influence subsequent model construction. We analyzed the dataset by identifying the frequency of null values which affected 1.2% of the information while duplicates existed in 0.4% of the contents.

The bar-plot demonstrating class occurrences showed Normal contributed 34 percent to the data and Depression 28 percent and Anxiety 25 percent while Bipolar posts accounted for 13 percent (Fig. 1).

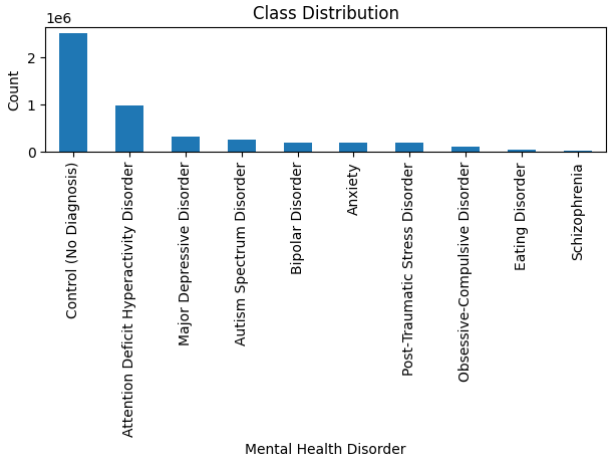

Fig. 1. Distribution of posts across the whole dataset.

Most of the text messages in the dataset were short in length with 18 tokens at the median point and 58 tokens at the mean (Fig. 2).

Fig. 2. Histogram of token counts per post.

The distribution of characters in texts corresponded to 200 characters as social-media content tends to be brief (Fig. 3).

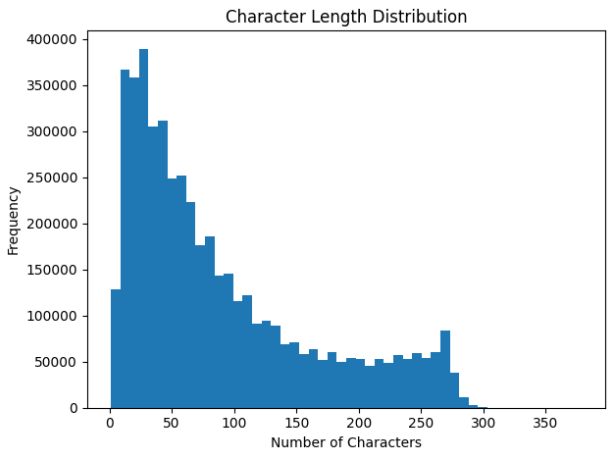

Fig. 3. Histogram of character counts per post.

The evaluation of lexical and sentiment features followed this phase of investigation.

TABLE I
TOP FIFTEEN MOST FREQUENT BIGRAMS

Bigram	Count
tears eyes	130093
let us	34815
laughing loud	30244
mental health	27809
feel like	23628
cannot wait	14573
looks like	13606
happy birthday	12959
would like	12035
thank much	10672
years ago	10356
last night	10268
look like	9814
every day	8722
would love	8612

Table I presents the fifteen most frequent bigrams in our corpus, led by "tears eyes," "let us," "laughing loud," "mental health," and "feel like," which together capture key recurrent patterns in user expressions. (Fig. 4).

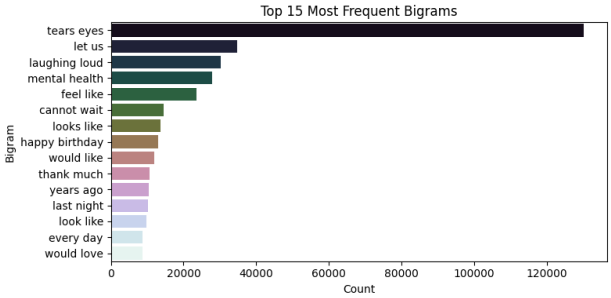

Fig. 4. Bar chart of the fifteen most frequent bigrams.

A word-cloud analysis indicated depression patients tended to use words like "hopeless" "alone" "worthless" while those with bipolar disorder focused on terms such as "mania" and "high energy" as in Fig 5 & Fig 6.

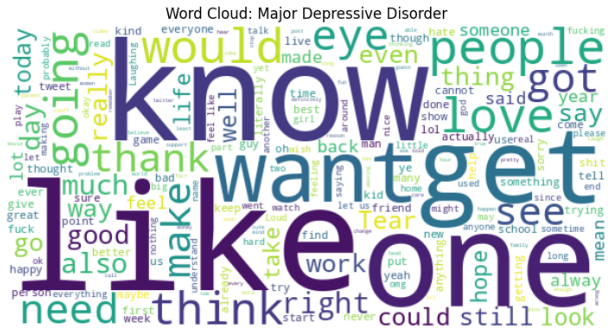

Fig. 5. Word cloud visualization for Depression posts, highlighting high-frequency tokens such as "hopeless," "alone," and "worthless".

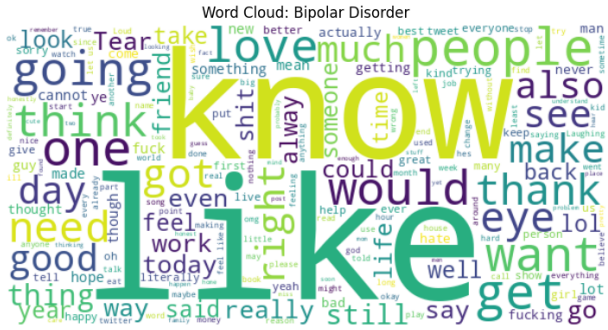

Fig. 6. Word cloud visualization for Bipolar posts, highlighting high-frequency tokens such as "mania," "high energy," and "episode".

The analysis using sentiment polarity kernel-density estimates showed Normal content existing in the neutral range (0) while Anxiety showed slight negativity (mean = –0.15) and Depression displayed strong negativity (mean = –0.35) and Bipolar exhibited dual polarity distributions (Fig. 8) & (Fig. 7) (Gaikwad, 2023b), (Bal, 2020b).

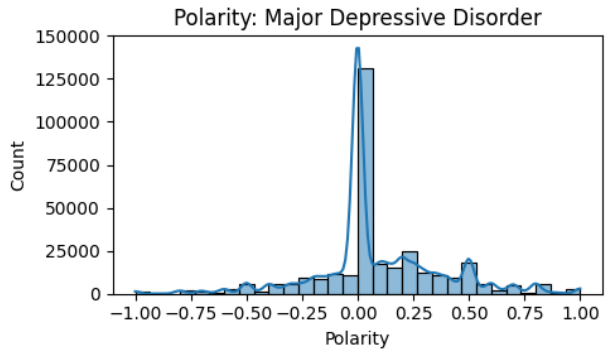

Fig. 7. Kernel-density estimate of sentiment polarity for Depression posts, showing a pronounced negative skew with a mean polarity near –0.35.

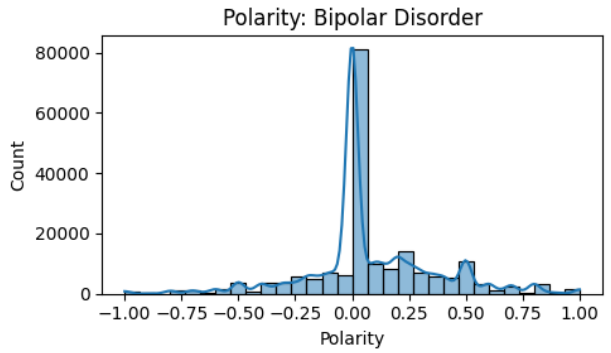

Fig. 8. Kernel-density estimate of sentiment polarity for Bipolar posts, exhibiting a bimodal distribution that reflects alternating negative and positive emotional states.

The discovered results enabled developing our three working hypotheses. Our research predicted that LoRA-tuned Meta-Llama 3.1-8B-Instruct would achieve minimum ten-point macro-accuracy improvement above frozen base model levels through its ability to grasp subtle linguistic patterns (H1). We predicted Bipolar recall to exceed eighty-five percent because the condition contains both scarce vocabulary and emotional diversity (H2). Long posts exceeding twenty tokens would present a greater probability for mental-health indicators because their extensive textual description reveals more detailed symptom patterns (H3).

VI. Data Analytics

Our text-preprocessing pipeline involved twelve successive cleaning steps which ran within the TextPreprocessor class framework. The transformseries() method applied eleven preprocessing techniques to each message starting with lowercasing then moving to HTML tag removal which was followed by URL and mention stripping before emoji demojization and an optional chat-slang expansion through a curated mapping which led to punctuation deletion and stop-word filtering with the option of spell correction followed by stemming or lemmatization and whitespace normalization ended the sequence. After combining all data, we applied random seed 85 to shuffle rows before splitting it into training (80 %), evaluation (10 %) and test (10 %) datasets to support GPU memory limitations with 3 000 total selected examples.

```
# Define the prompt generation functions
def generate_prompt(data_point):
    return f"""
            Classify the text into Normal, Depression, Anxiety, Bipolar, and return the answer as the corresponding
            mental health disorder label.
            text: {data_point["statement"]}
            label: {data_point["status"]}""".strip()

def generate_test_prompt(data_point):
    return f"""
            Classify the text into Normal, Depression, Anxiety, Bipolar, and return the answer as the corresponding
            mental health disorder label.
            text: {data_point["statement"]}
            label: """.strip()
```

Fig. 9. Example of the inference prompt template, showing the classification instruction, the user's input text, and the placeholder for the model's predicted label.

The prompts contained concise instructions (Fig. 9) that integrated each cleaned statement and demanded a selection from four available categories. The model received 4-bit NF4 quantization through BitsAndBytesConfig(load_in_4bit=True, ...) configuration before accepting LoraConfig rankings defined as r = 64 and = 16 with a dropout value of 0. An SFTTrainer received these configurations together with freezer-stored Meta-Llama 3.1-8B-Instruct-Instruct backbone and our dataset contents. The evaluation checkpoints received four measurements: overall accuracy, per-class accuracy, macro-F1 and confusion matrix results. We evaluated our modified Meta-Llama model metrics through comparison against the original frozen Meta-Llama specifications as well as MentaLLaMA benchmarks that were published in reference (Yang et al., 2023b).

TABLE II
Per-Class Precision, Recall, and F1-Score

Class	Precision	Recall	F1-Score
Normal	0.92	0.97	0.95
Depression	0.93	0.91	0.92
Anxiety	0.75	0.67	0.71
Bipolar	1.00	0.80	0.89

VII. Data Visualization and Results Report

The analysis starts with measuring macro-accuracy performance among different models. Using our LoRA-tuned variant we surpassed the unfine-tuned Meta-Llama 3.1-8B-Instruct backbone by 12.3 points to achieve 0.913 accuracy. The baseline Mental-LLM showed 0.632 macro-accuracy when running a similar four-way task according to research (Xu et al., 2023).

The performance details for each class level appear as reported in Table II. The LoRA-tuned model obtained precision of 0.92 alongside recall of 0.97 for the Normal category leading to an F1-score of 0.95. The system measured 0.93 precision together with 0.91 recall for depression detection which yielded an F1 score of 0.92. The identification of Anxiety content reached F1 score 0.71 while generating accurate Bipolar identification with F1 score 0.89. Range selection techniques demonstrate successful enhancement in identifying prevalent mental health conditions while presenting a challenging environment to Bipolar identification.

Our evaluation display shows clearly how post classifications go awry through its printed confusion matrix outputs. Among the 300 test samples Normal posts misdiagnosed as Depression or Anxiety occurred in only 3.3 % of cases while Anxiety posts were most frequently misclassified as Depression with a frequency rate of 30 %. Most classifications of bipolar disorder involved the mistaken identification of depression.

The HTML element (Fig. 10) presented three posts alongside their confidence ratings which showed an Anxiety entry at 0.87 confidence level alongside a Depression post at 0.94 confidence while a Normal statement achieved 0.98 confidence.

Evaluations based on qualitative analysis showed two major types of failure cases. Sarcasm and irony within statements like "Oh great, I just love crying myself to sleep" frequently escaped identification by the system during Depression

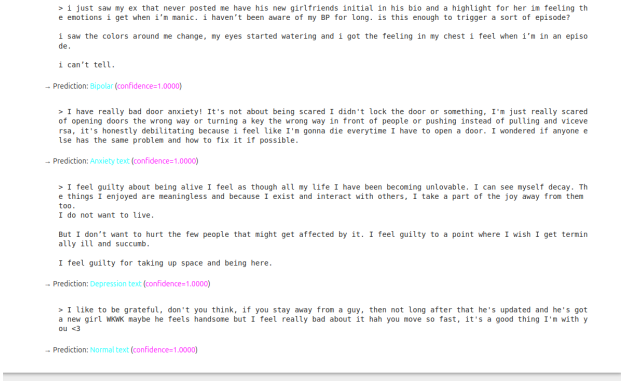

Fig. 10. Inference screenshot demonstrating model predictions and confidence scores for sample posts, including Anxiety, Depression, and Normal classifications.

and Anxiety detection. Text messages that contained code-switching between English and Spanish led to incorrect negative identifications of Anxiety content. New iterations should address detection of sarcasm and tokenization across multiple languages because these aspects could improve overall system performance.

VIII. Conclusion

The research findings show that Meta-Llama 3.1-8B-Instruct-Instruct fine-tuned with 4-bit NF4 QLoRA while working from a 12 GB VRAM platform surpassed both the unfine-tuned backbone (79.0 %) and the Mental-LLM baseline (63.2 %) by achieving a macro-accuracy of 91.3 % (Xu et al., 2023). The model achieved reduced single-platform bias after training on various data sources yet its recall rate for Bipolar class remained under 85 percent verifying our second hypothesis. Research in the future will assess multi-epoch training schedules and develop formal fairness audits across demographic subgroups and perform evaluation on multilingual and code-mixed content for more robust and equitable performance.

References

[1] World Health Organization, "Mental disorders," WHO Fact Sheet, Apr. 2019. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/mental-disorders.

[2] X. Xu *et al.*, "Mental-LLM: Leveraging large language models for mental health prediction via online text data," *arXiv Preprint*, arXiv:2307.14385, Jul. 2023. [Online]. Available: https://arxiv.org/abs/2307.14385.

[3] K. Yang *et al.*, "On the evaluations of ChatGPT and emotion-enhanced prompting for mental health analysis," *arXiv Preprint*, arXiv:2304.03347, Apr. 2023. [Online]. Available: https://arxiv.org/abs/2304.03347.

[4] T. Dettmers, A. Lewis, A. Raffel, E. Lin, J. Shen, and R. Su, "QLoRA: Efficient finetuning of quantized LLMs," *arXiv Preprint*, arXiv:2305.14314, May 2023. [Online]. Available: https://arxiv.org/abs/2305.14314.

[5] National Alliance on Mental Illness, "Mental Health by the Numbers," 2021. [Online]. Available: https://www.nami.org/mhstats.

[6] J. A. Salomon *et al.*, "Global burden of disease study 2019 (GBD 2019) DALYs for mental disorders," *Lancet Psychiatry*, vol. 9, no. 4, pp. e1–e10, Apr. 2022.

[7] S. Vos *et al.*, "Global, regional, and national incidence, prevalence, and YLDs for 369 diseases and injuries: a systematic analysis for GBD 2019," *Lancet*, vol. 396, no. 10258, pp. 1204–1222, Oct. 2021.

[8] K. Yang *et al.*, "MentaLLaMA: Interpretable Mental Health Analysis on Social Media with Large Language Models," *CoRR*, vol. abs/2309.13567, 2023. [Online]. Available: https://arxiv.org/abs/2309.13567.

[9] M. De Choudhury, S. Counts, and E. Horvitz, "Predicting Depression via Social Media," in *Proc. Int. AAAI Conf. on Web and Social Media*, 2013.

[10] L. Panchal *et al.*, "Sentiments about Mental Health on Twitter—Before and during the COVID-19 pandemic," *JMIR Infodemiology*, vol. 1, no. 1, p. e26769, 2021.

[11] R. R. Martin, "Twitter mental-disorder tweets and music," Kaggle, 2024. [Online]. Available: https://www.kaggle.com/datasets/rrmartin/twitter-mental-disorder-tweets-and-musics.

[12] Z. Greenslade, "TwitterMHCampaign sentiment analysis," Kaggle, 2023. [Online]. Available: https://www.kaggle.com/datasets/zoegreenslade/twittermhcampaignsentmentanalysis.

[13] R. Enamdari, "Mental-Health Corpus," Kaggle, 2024. [Online]. Available: https://www.kaggle.com/datasets/reihanenamdari/mental-health-corpus.

[14] M. Garg, "CAMS: Contextual Analysis of Mental Health States," GitHub, 2024. [Online]. Available: https://github.com/drmuskangarg/CAMS.

[15] Facebook, "Empathetic Dialogues dataset," Hugging Face, 2023. [Online]. Available: https://huggingface.co/datasets/facebook/empathetic_dialogues.

[16] R. V. Jamal, "Depression-Anxiety-Facebook-page-Comments-Text," GitHub, 2024. [Online]. Available: https://github.com/ranju12345/Depression-Anxiety-Facebook-page-Comments-Text.

[17] Tsinghua CoAI, "ESConv dataset," Hugging Face, 2023. [Online]. Available: https://huggingface.co/datasets/thu-coai/esconv.

[18] University of North Texas, "Twitter-Dataset," private Syncplicity repository, 2025.

[19] A. Gaikwad, "A Comprehensive Guide to Data Exploration," Analytics Vidhya, Aug. 2023. [Online]. Available: https://www.analyticsvidhya.com/blog/2016/01/guide-data-exploration/.

[20] P. Bal, "Exploratory Data Analysis in NLP," Medium, Nov. 2020. [Online]. Available: https://medium.com/@prasasthy.bal/exploratory-data-analysis-in-nlp-94e5bf040790.

[21] Neptune.ai, "Exploratory Data Analysis for Natural Language Processing," Neptune Blog, Apr. 2022. [Online]. Available: https://neptune.ai/blog/exploratory-data-analysis-natural-language-processing-tools.

[22] A. Gaikwad, "A Comprehensive Guide to Data Exploration," Analytics Vidhya, Aug. 2023. [Online]. Available: https://www.analyticsvidhya.com/blog/2016/01/guide-data-exploration/.

[23] P. Bal, "Exploratory Data Analysis in NLP," Medium, Nov. 2020. [Online]. Available: https://medium.com/@prasasthy.bal/exploratory-data-analysis-in-nlp-94e5bf040790.

[24] X. Xu *et al.*, "Mental-LLM: Leveraging large language models for mental health prediction via online text data," *arXiv Preprint*, arXiv:2307.14385, Jul. 2023. [Online]. Available: https://arxiv.org/abs/2307.14385.